

# Redrob Candidate Intelligence Ranker

An end-to-end, offline, explainable ranking system for selecting the top 100 candidates from 100,000 profiles.

## One Solution, Not Separate Versions

The final system is a single pipeline: understand the JD, extract evidence from each profile, score candidate fit, learn from reviewed labels, apply safety checks, generate deterministic reasoning, and validate the final CSV.

| Validator               | Top-100 fake risk > 0.10 | Under 5 yrs   | Pair accuracy       |
|-------------------------|--------------------------|---------------|---------------------|
| Passed                  | 0                        | 0             | 0.961               |
| Submission CSV is valid | max risk 0.08            | min years 5.0 | learned label order |

## 1. Problem Understanding

- The task is not generic resume matching. It is a relevance-ranking problem for a specific ML/search/recommendation role.
- The system must identify candidates who can build production ranking, retrieval, recommendation, semantic search, and evaluation systems.
- It must also avoid fake-looking or low-quality profiles, because a keyword-heavy profile can look impressive while being unreliable.

## 2. Key JD Requirements Extracted

- Strong ML/search/recommendation background.
- 5 to 9 years is the ideal experience band for this role.
- Production ownership matters more than research-only or keyword-only experience.
- Ranking evaluation evidence is highly valuable: NDCG, MRR, recall@K, offline-to-online correlation, A/B testing, and human relevance labels.
- Practical hiring signals matter: India or willing to relocate, response rate, notice period, verified contact signals, and recent activity.

### 3. Candidate Signals Used

- Profile signals: title, current company, industry, location, years of experience, headline, and summary.
- Career signals: job titles, company history, duration, production projects, ranking/retrieval ownership, evaluation language, and scale.
- Skill signals: information retrieval, learning-to-rank, recommendation systems, BM25, FAISS, vector databases, embeddings, semantic search, MLflow, MLOps, and LLM/RAG where relevant.
- Platform signals: profile completeness, verified email/phone, LinkedIn connected, recruiter response rate, open-to-work flag, notice period, GitHub activity, saved-by-recruiters, and interview completion.
- Risk signals: mismatched years, suspicious timelines, inflated summaries, domain mismatch, stale profile behavior, and fake-risk score.

### 4. Ranking Methodology

- First, every candidate is converted into structured features. These features capture technical fit, career proof, logistics, and profile-risk.
- Second, a reviewed training file with 520 labeled examples teaches a small pairwise linear reranker how to order candidates.
- The reranker is trained with pairwise logistic loss. In simple terms, it learns that a candidate labeled 5 should score above a candidate labeled 3, and a candidate labeled 3 should score above a candidate labeled 1.
- The final model blends learned ranking behavior with the original feature score as a guardrail, so the model cannot drift too far from the problem-specific rules.
- No live API or cloud model is needed at ranking time. The final run is CPU-only and offline.

### 5. Label Strategy

- Two independent LLM-generated label files were used only as offline labeling assistance.
- A consensus file was created, then the high-disagreement rows were reviewed and corrected.
- Final training data has 520 examples. The reviewed disagreement set corrected cases where one model was too strict or too generous.
- Review result: 18 upgrades, 16 downgrades, and 24 kept as-is.
- The final ranking system does not call an LLM. Labels are training data only.

### 6. Fake Profile and Low-Quality Profile Handling

- The system checks whether structured years, career timeline, and summary claims are consistent.
- It penalizes suspicious profiles where career duration does not match stated experience.
- It also penalizes profiles that are keyword-rich but lack real ownership of ranking, retrieval, recommendation, or evaluation work.
- Profiles with high fake-risk are kept out of the final top 100. QA confirms zero top-100 candidates above fake-risk 0.10.

## 7. Reasoning Generation

- Reasoning is deterministic and template-based. It is not generated by an LLM.
- For each selected candidate, the code writes a short explanation from actual fields: years, title, company, evidence hits, location, notice period, response rate, and evaluation signal.
- This prevents hallucination because the reasoning only uses evidence already detected in the candidate profile.

## End-to-End Workflow

|                     |                 |                    |                 |                   |                 |             |
|---------------------|-----------------|--------------------|-----------------|-------------------|-----------------|-------------|
| JD + challenge docs | Candidate JSONL | Feature extraction | Reviewed labels | Pairwise reranker | Guardrails + QA | Top-100 CSV |
|---------------------|-----------------|--------------------|-----------------|-------------------|-----------------|-------------|

- Read job description and challenge docs.
- Stream candidates from JSONL without loading the full 465 MB file into memory.
- Normalize text from profile, skills, career history, and platform signals.
- Detect evidence groups: title fit, retrieval, ranking, evaluation, production, relevance systems, skills, logistics, and risk.
- Score and featurize every candidate.
- Train an offline pairwise reranker from reviewed labels.
- Rank all candidates, write top 100, and generate deterministic reasoning.
- Run validator and QA: row count, fake-risk leakage, experience range, target-title coverage, and reasoning uniqueness.
- Package final CSV, audit, model, training data, QA report, code, and docs.

## Top Ranked Evidence Snapshot

| Rank | Candidate    | Profile                                                                        | Evidence                                                                                                                                                                                                                        |
|------|--------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | CAND_0030468 | 5.4 yrs Senior Applied Scientist<br>Swiggy - Indore, Madhya Pradesh            | connect users with relevant information , relevant information at scale , ranking and retrieval systems , decide what to show , search & discovery , offline metrics , online numbers , offline experimentation                 |
| 2    | CAND_0006567 | 7.9 yrs Senior AI Engineer<br>Meta - Noida, Uttar Pradesh                      | connect users with relevant information , relevant information at scale , ranking and retrieval systems , decide what to show , search and discovery experience , discovery experience , most relevant results , user's intent  |
| 3    | CAND_0011687 | 7.8 yrs Senior NLP Engineer<br>Niramai - Indore, Madhya Pradesh                | offline metrics , intelligence layer end-to-end , hybrid retrieval , semantic search , retrieval , embedding , embeddings , faiss                                                                                               |
| 4    | CAND_0080766 | 8.8 yrs Staff Machine Learning Engineer<br>Salesforce - Coimbatore, Tamil Nadu | connect users with relevant information , relevant information at scale , ranking and retrieval systems , decide what to show , surface the right thing , right thing at the right time , millions of items , millions of users |
| 5    | CAND_0037980 | 9.0 yrs Senior Applied Scientist<br>Niramai - Kolkata, West Bengal             | connect users with relevant information , relevant information at scale , ranking and retrieval systems , decide what to show , surface the right thing , right thing at the right time , millions of items , millions of users |

## Results and Quality Checks

- Official validator passed.
- Top 100 count: 100.
- Experience range: 5.0 to 9.0 years, median 6.7.
- Fake-risk max: 0.08; candidates above 0.10: 0.
- Reasoning unique count: 100.
- Validation pair accuracy: 0.961.
- Validation MAP for labels  $\geq 3$ : 0.976.

## What Makes This Strong for a National Hackathon

- It is problem-specific, not a generic resume keyword matcher.
- It combines interpretable feature engineering with a learned reranker.
- It has a clear fake-profile strategy.
- It produces auditable explanations without hallucination.
- It is reproducible, offline, CPU-only, and fits the submission constraints.
- It includes a complete submission package with final CSV, QA report, training labels, model, code, docs, and audit trail.

## Known Risk and Final Guardrail

The current QA confirms strong technical quality. One practical guardrail still worth applying before final upload is a hard India-or-relocation filter, because QA does not currently flag non-India candidates who are not willing to relocate. This is not a new feature; it is a hiring-practicality guardrail.